

1) Need of Flowchart

A flowchart is a graphical representation of an algorithm. It helps in understanding logic easily, improves communication, helps in debugging, and provides proper documentation.

2) Pre-increment and Post-increment

Pre-increment (++i) increases the value before use, while post-increment (i++) uses the value first and then increases it.

```
int a=5;
int b=++a; // a=6, b=6

int x=5;
int y=x++; // y=5, x=6
```

3) Expression Evaluation

Given a=5, b=10, c=a+b*2. According to operator precedence, multiplication happens first.

b*2 = 20, then a+20 = 25. So c = 25.

4) printf and scanf

printf() is used to display output, scanf() is used to take input.

```
#include <stdio.h>
int main(){
    int a;
    printf("Enter: ");
    scanf("%d",&a);
    printf("%d",a);
    return 0;
}
```

5) Pattern Program

```
#include <stdio.h>
int main(){
    for(int i=1;i<=8;i++){
        printf("* ");
    }
    return 0;
}
```

6) Format Specifiers

%d for integer, %f for float, %c for character, %s for string.

7) Largest of Three Numbers

```
#include <stdio.h>
```

```

int main(){
    int a,b,c;
    scanf("%d%d%d",&a,&b,&c);
    if(a>=b && a>=c)
        printf("%d",a);
    else if(b>=a && b>=c)
        printf("%d",b);
    else
        printf("%d",c);
}

```

8) Grade using switch

```

#include <stdio.h>
int main(){
    int m;
    scanf("%d",&m);
    switch(m/10){
        case 10: case 9: case 8: printf("A"); break;
        case 7: printf("B"); break;
        case 6: printf("C"); break;
        case 5: printf("D"); break;
        default: printf("F");
    }
}

```

9) Arithmetic Operations using nested if

```

#include <stdio.h>
int main(){
    int a,b,ch;
    scanf("%d%d",&a,&b);
    scanf("%d",&ch);
    if(ch==1) printf("%d",a+b);
    else if(ch==2) printf("%d",a-b);
    else if(ch==3) printf("%d",a*b);
    else if(ch==4){
        if(b!=0) printf("%d",a/b);
        else printf("Error");
    }
}

```

OR: Grade using if-else

```

#include <stdio.h>
int main(){
    int m;
    scanf("%d",&m);
    if(m<0 || m>100) printf("Invalid");
    else if(m>=80) printf("A");
    else if(m>=70) printf("B");
    else if(m>=60) printf("C");
    else if(m>=50) printf("D");
    else printf("F");
}

```